

Module 2: Infrastructure Security



Topic 3: Vulnerability Assessment and Penetration Testing



Vulnerability Assessment

A vulnerability assessment is the process through which we identify the weak points that could be exploited on your network, along with how important the associated threat really is. Silent Breach conducts a thorough analysis to determine the attack surface that is intentionally or unintentionally exposed, and correlates it with a risk value to determine your security posture.



Vulnerability Assessment

- Tools:
 - OpenVAS
 - **Nessus**
 - Nexpose
 - Outpost24
 - Qualys

Vulnerability Assessment

Nessus

win7 / 10.0.0.22

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Configure

Audit Trail

Launch

Report

Export

Vulnerabilities

17

Filter

Search Vulnerabilities



17 Vulnerabilities

☐	Sev	Name	Family	Count		
☐	MIXED	3 Microsoft Windows (Multiple Issues)	Windows	3		
☐	MEDIUM	SMB Signing not required	Misc.	1		
☐	INFO	7 SMB (Multiple Issues)	Windows	8		
☐	INFO	DCE Services Enumeration	Windows	7		
☐	INFO	Nessus SYN scanner	Port scanners	3		
☐	INFO	Common Platform Enumeration (CPE)	General	1		
☐	INFO	Device Type	General	1		
☐	INFO	Ethernet Card Manufacturer Detection	Misc.	1		
☐	INFO	Ethernet MAC Addresses	General	1		
☐	INFO	Local Checks Not Enabled (info)	Settings	1		
☐	INFO	Nessus Scan Information	Settings	1		
☐	INFO	Nessus Windows Scan Not Performed with Admin Privileges	Settings	1		

Host Details

IP: 10.0.0.22
MAC: 08:00:27:EA:A6:49
OS: Microsoft Windows 7 Ultimate
Start: Today at 4:19 AM
End: Today at 4:32 AM
Elapsed: 13 minutes
KB: [Download](#)

Vulnerabilities



● Critical
● High
● Medium
● Low
● Info

Vulnerability Assessment

Nessus

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

TENABLE

Community

Research

Tenable News

Dell EMC OpenManage
Server Administrator
Authentic...

Read More

win7 / 10.0.0.22 / Microsoft Windows (Multiple Issues)

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Vulnerabilities 17

Search Vulnerabilities

3 Vulnerabilities

Sev	Name	Family	Count		
☐ CRITICAL	Unsupported Windows OS (remote)	Windows	1		
☐ HIGH	MS17-010: Security Update for Microsoft Windows SMB Server (4013389) (ETERNALBLUE) (ETERNAL...	Windows	1		
☐ INFO	WMI Not Available	Windows	1		

Scan Details

Policy:

Advanced Scan

Status:

Completed

Scanner:

Local Scanner

Start:

Today at 4:17 AM

End:

Today at 4:32 AM

Elapsed:

15 minutes

Vulnerabilities

Critical

High

Medium

Low

Info

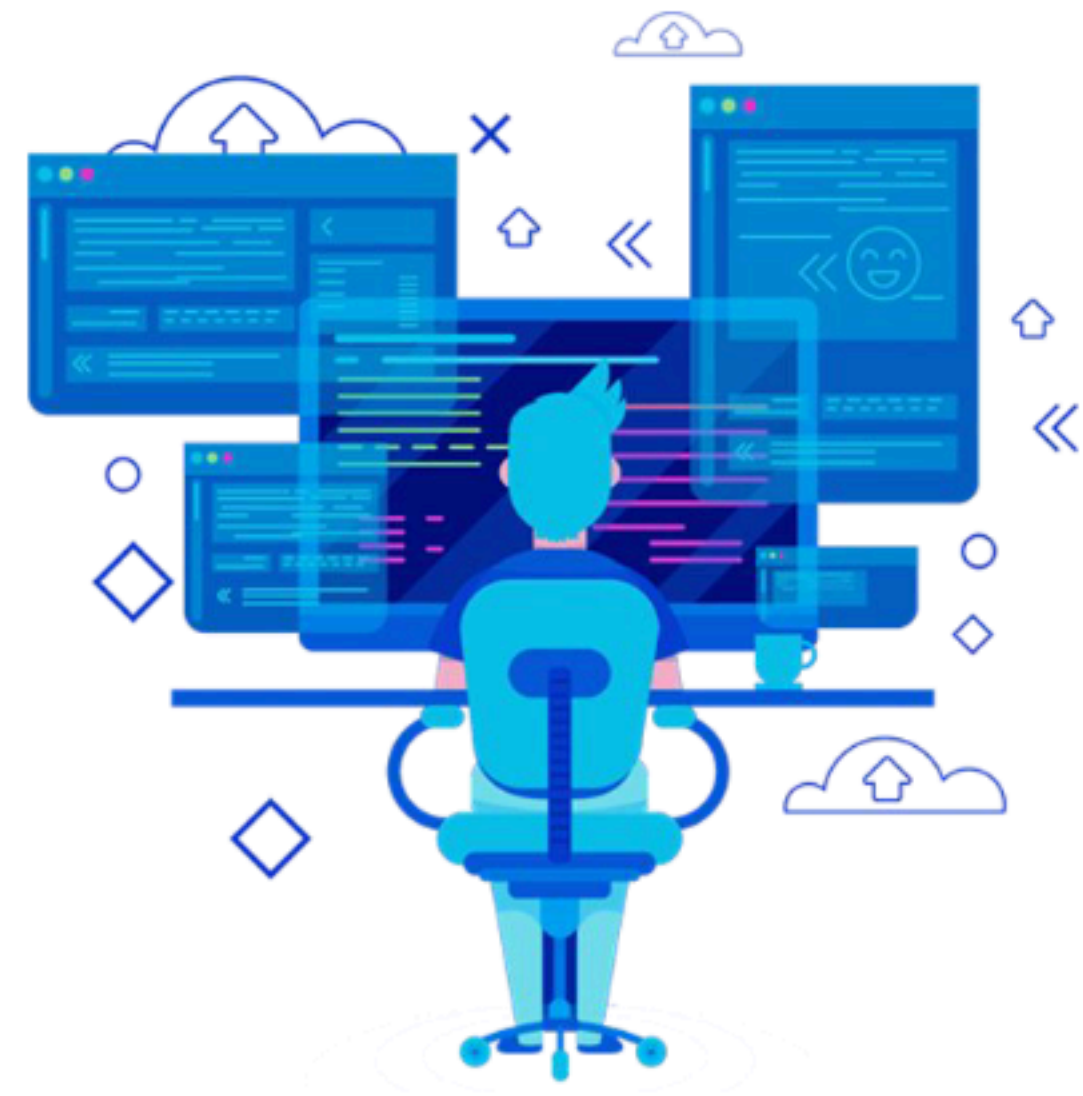
Vulnerability Assessment

Reporting

High Level Report	Vulnerability Report	Step by Step Guide
Statistics	Each IP with Vulnerability	Starting with NMAP how to Scan
Overview of Network	Contain Exploit Details	Trail of Thoughts what we were thinking
How Secure the network	Screenshot of Successful Exploit	Fail Exploits
Works Well for Management	Screenshot of Privilege Escalation	Trial and Errors
	How To Fix the Vulnerability	

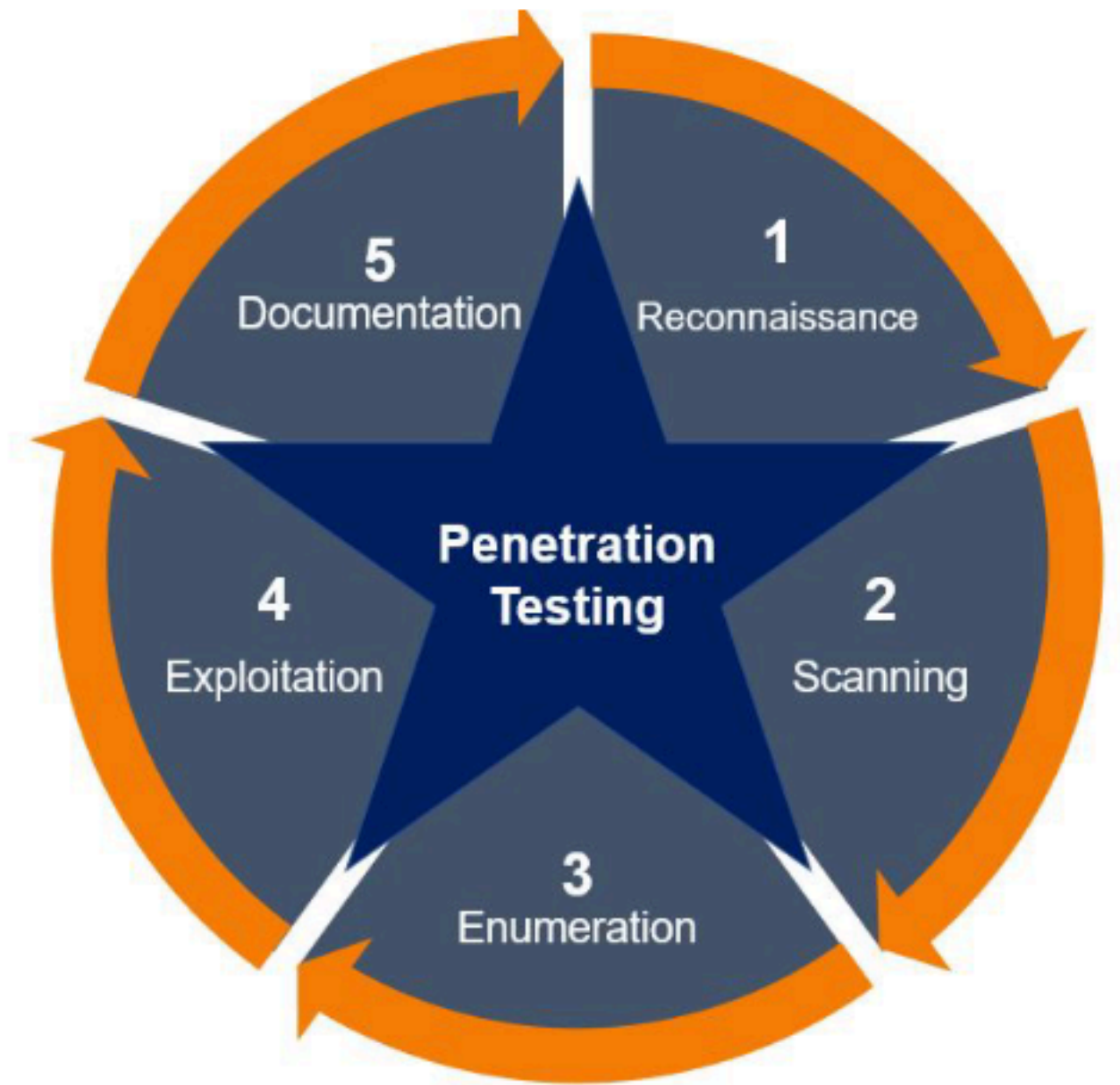
Introduction to Penetration Testing

Penetration testing (also called **pen testing**) is the practice of **testing** a computer system, network or Web application to find vulnerabilities that an attacker could exploit.



Introduction to Penetration Testing

- Information Gathering / Reconnaissance
- Scanning
- Enumeration
- Exploitation
- Documentation and reporting



Types of Penetration Testing



Black Box

- Zero knowledge
- No access /credentials given
- Test as external attacker



Gray Box

- Partial knowledge
- Minimum access or credentials given
- Test as normal user



White Box

- Full knowledge
- All access and credentials given
- Test as developer

Why Penetration Testing?

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- 1- Find vulnerable systems before hackers do.
- 2- Determine strength of security controls used in environment.
- 3- Prioritize security investments and security controls needed.
- 4- Adhere to some regulatory or framework requirements.
- 5- Protect corporate image and brand reputation.

Vulnerability Assessment Vs Penetration Testing

Vulnerability Assessment	Penetration testing
Automated activity	Manual Activity
Low price (price dependent on used tool)	Expensive (depending on experience of pentester)
Takes few time to be completed	Takes long time
Can be done on large scale	Limited scope
Detect vulnerabilities without exploitation	Exploit available vulnerabilities
Multiple false positives	Rare false positives